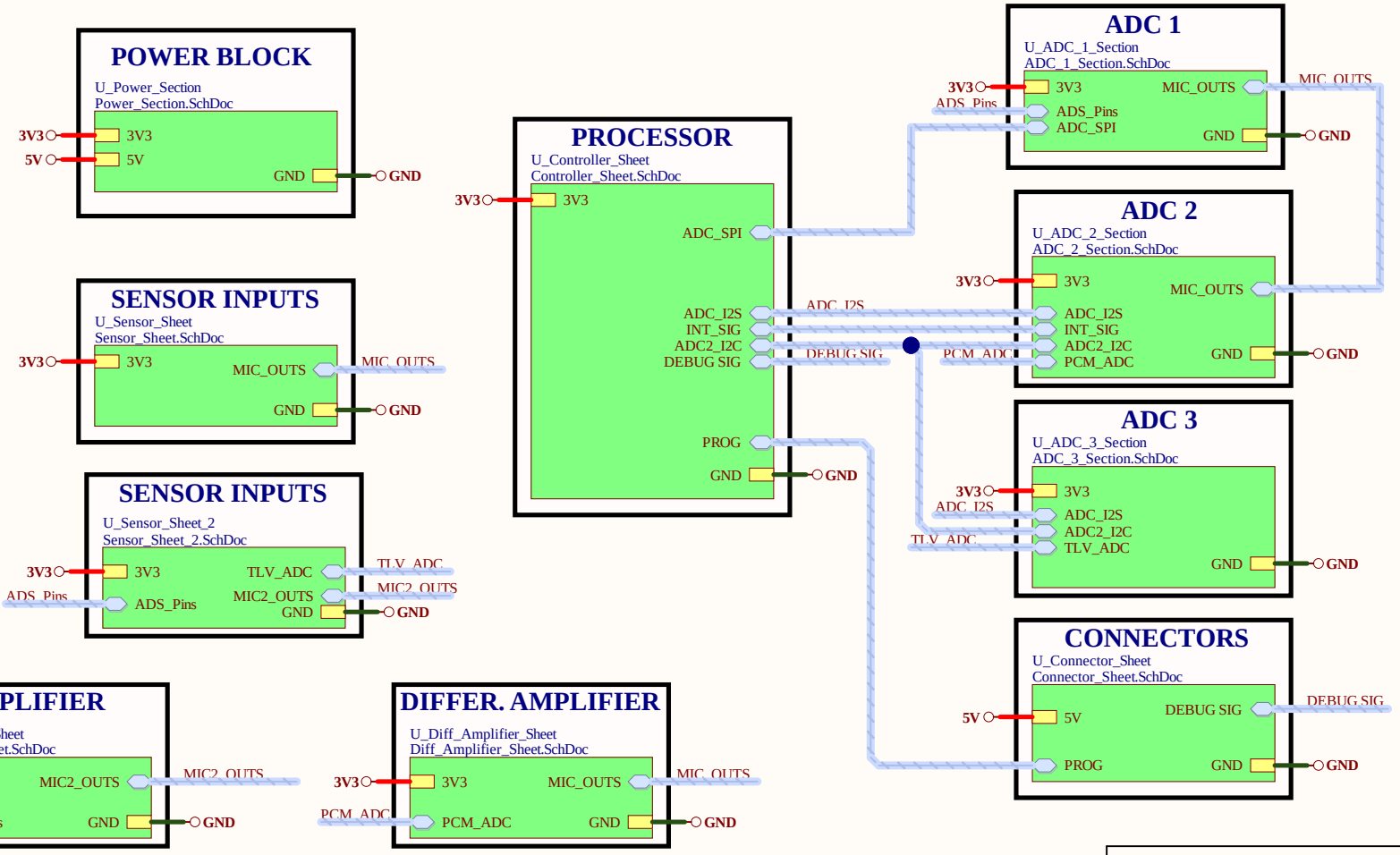
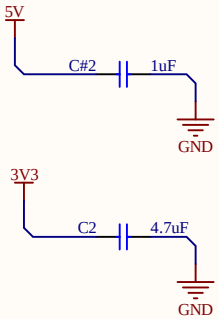


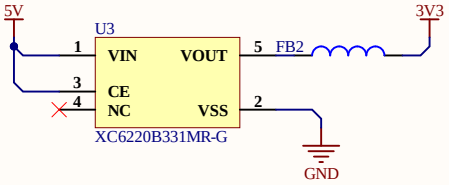
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Title: Top_Sheet.SchDoc	
File: Sound_Sensor.PrjPcb	
Date: 2/7/2023	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 11

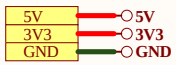
DECOUPLING CAPS



3.3V LDO



Fixed 3.3V output
150mV dropout @500mA
Max output current is 1A
Enable High to activitate the chip

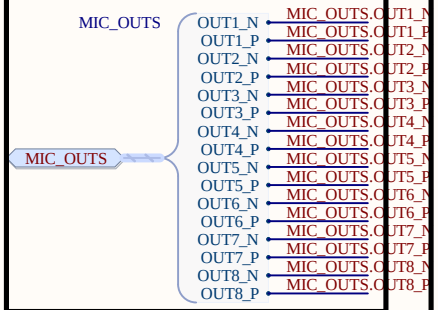


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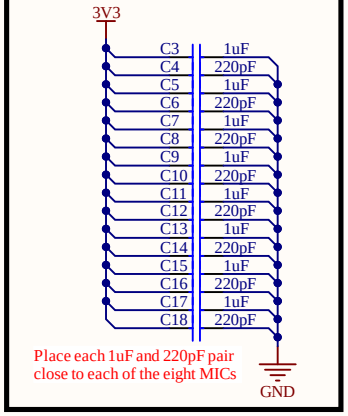
Title: Power_Section.SchDoc	
File: Sound_Sensor.PrjPcb	
Date: 2/7/2023	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 11

Set MIC differential traces to 10mils with 10mils clearnace between the two differential traces

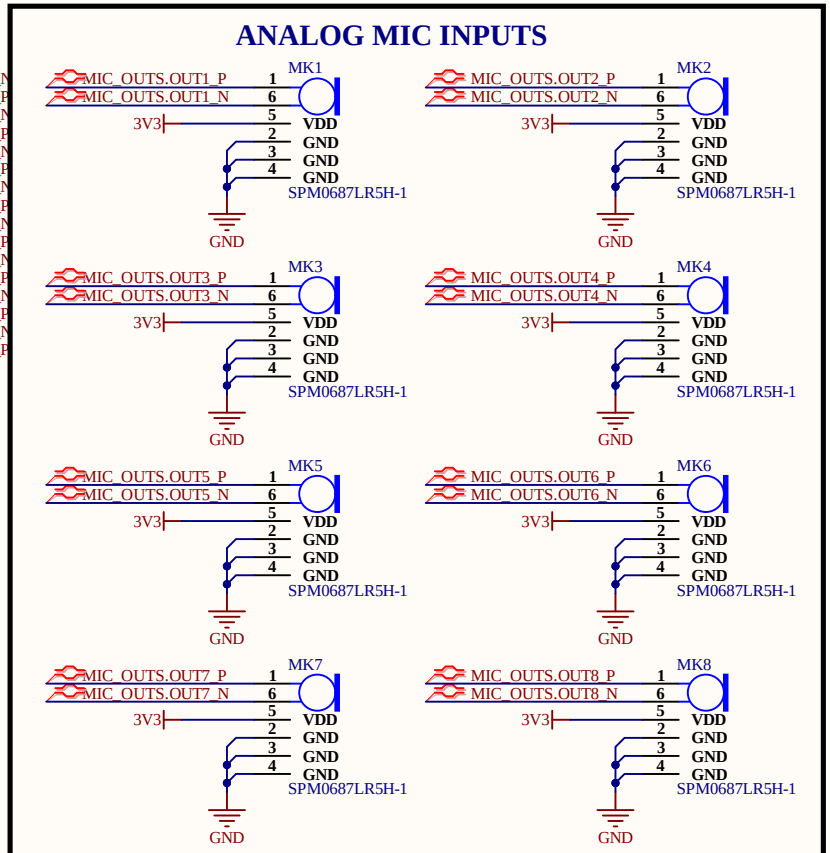
MIC OUT SIGNALS



DECOUPLING CAPS



ANALOG MIC INPUTS

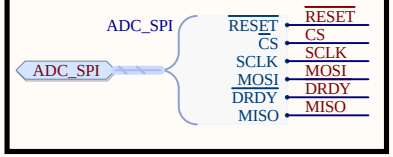




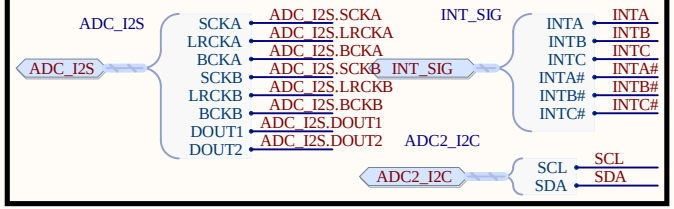
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Title: Sensor_Sheet.SchDoc	
File: Sound_Sensor.PrjPcb	
Date: 2/7/2023	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 11

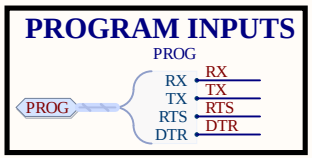
ADC1 OUTPUT



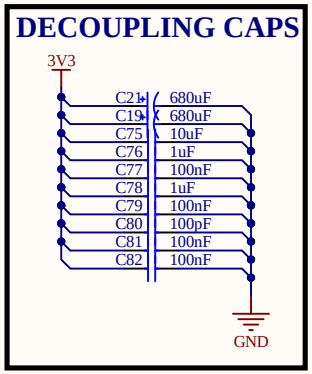
ADC 2 OUTPUT SIGNALS



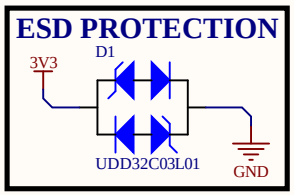
PROGRAM INPUTS



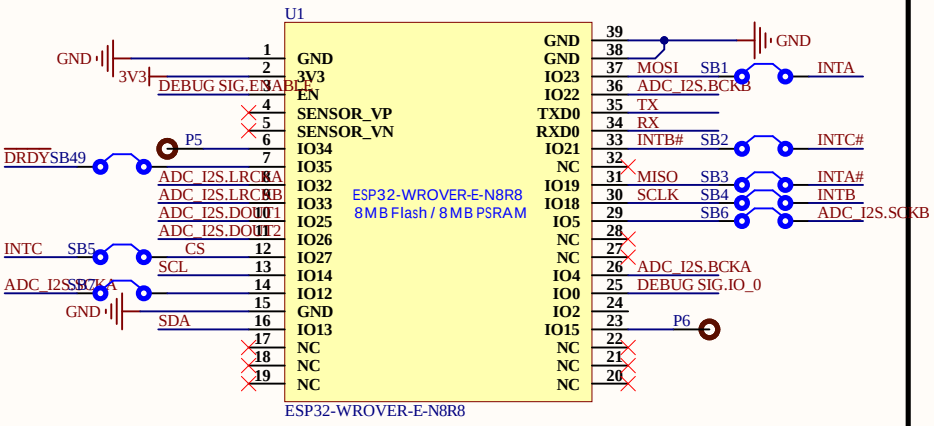
DECOUPLING CAPS



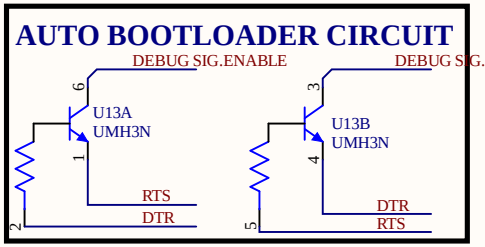
ESD PROTECTION



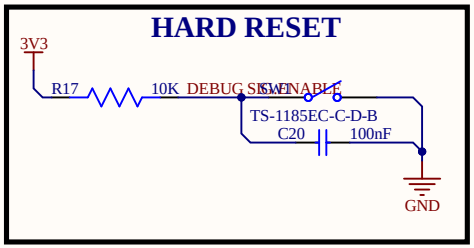
ESP32 Module



AUTO BOOTLOADER CIRCUIT



HARD RESET



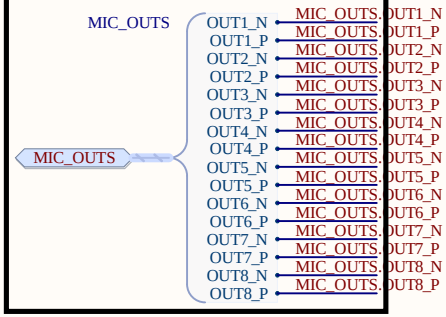




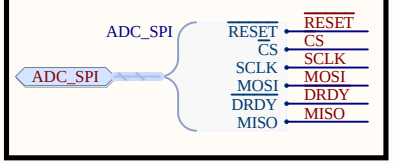
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D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg.
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Title: Controller_Sheet.SchDoc	
File: Sound_Sensor.PrjPcb	
Date: 2/7/2023	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 11

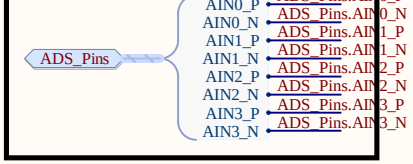
MIC OUT SIGNALS



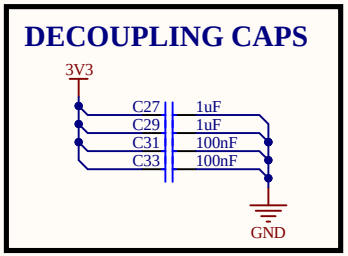
ADC OUTPUT



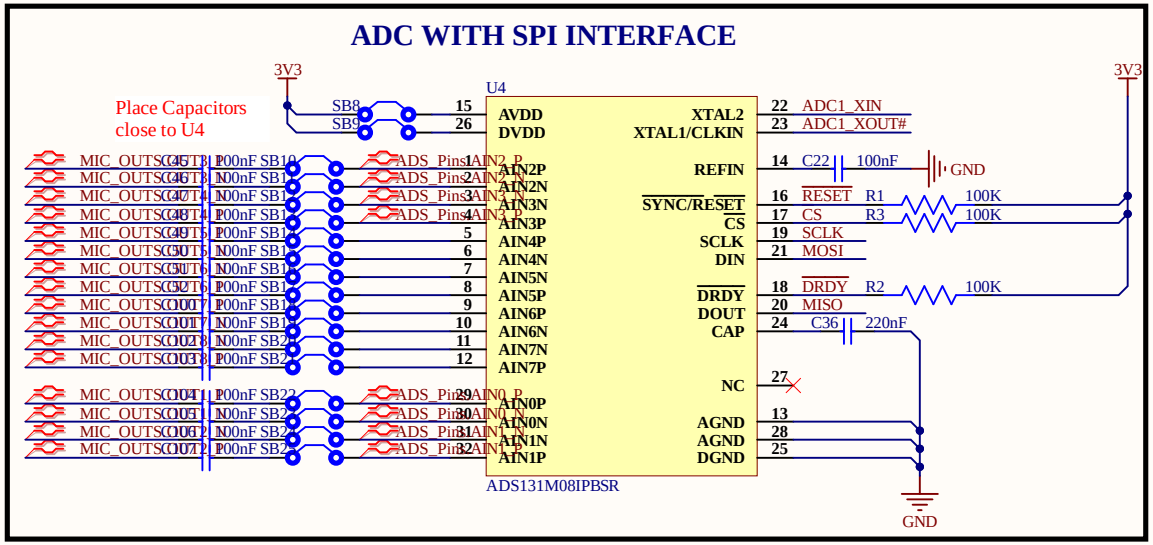
ADS131 SIGNALS



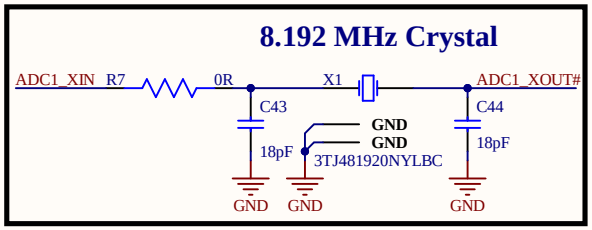
DECOUPLING CAPS



ADC WITH SPI INTERFACE



8.192 MHz Crystal





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Title: ADC_1_Section.SchDoc	
File: Sound_Sensor.PrjPcb	
Date: 2/7/2023	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 11

I2C CONTROL SIGNALS

ADC2_I2C

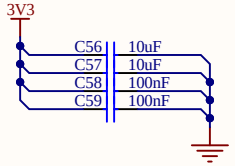
ADC2_I2C — SCL — SCL; SDA — SDA

MIC OUT SIGNALS

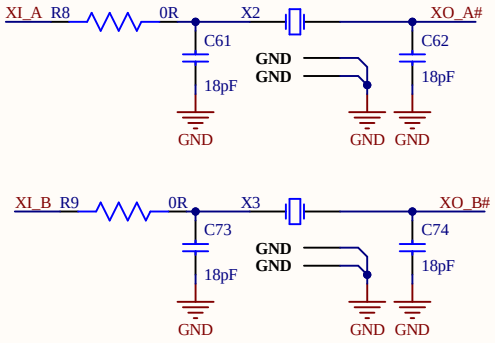
MIC_OUTS

MIC_OUTS — OUT1_N … OUT8_N, OUT1_P … OUT8_P — MIC_OUTS.OUT1_N … MIC_OUTS.OUT8_N, MIC_OUTS.OUT1_P … MIC_OUTS.OUT8_P

DECOUPLING CAPS



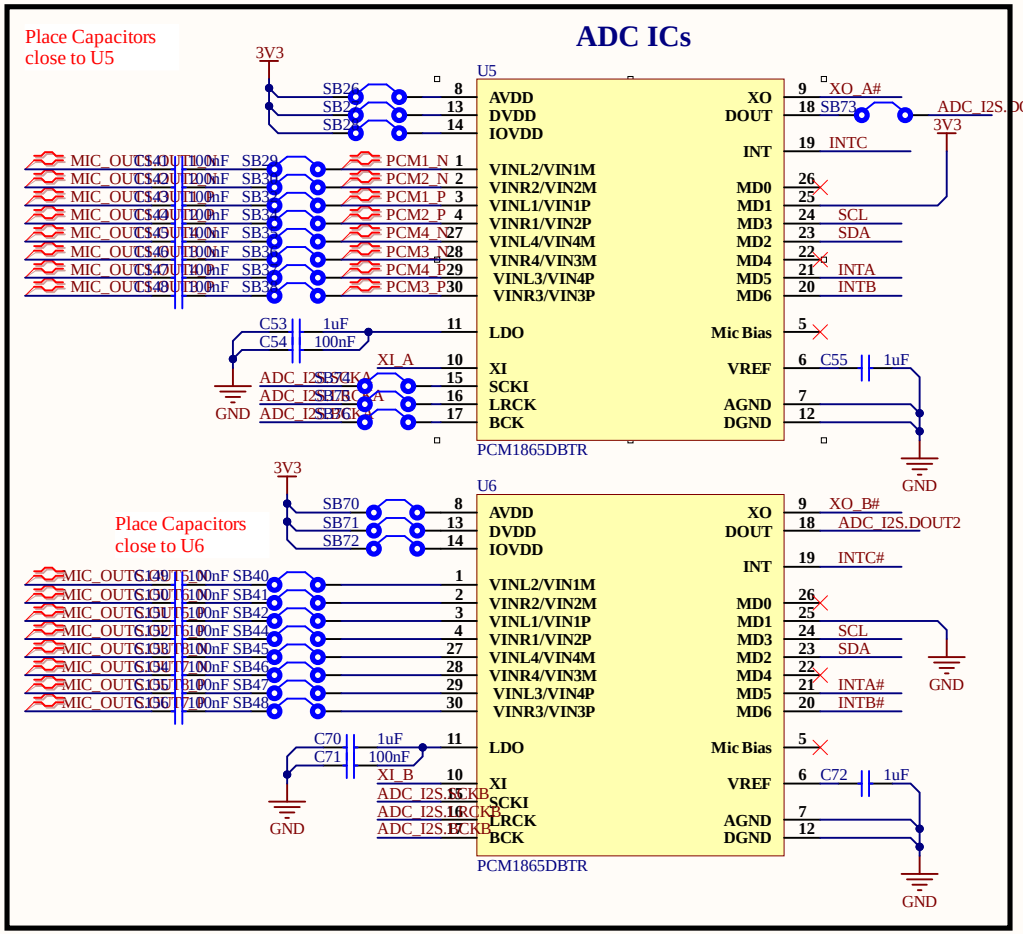
24.576 MHz Crystals



ADC ICs

Place Capacitors close to U5

Place Capacitors close to U6





PCM_ADC — PCM1_N, PCM2_N, PCM1_P, PCM2_P, PCM4_N, PCM3_N, PCM4_P, PCM3_P

ADC_I2S — SCKA, LRCKA, BCKA, SCKB, LRCKB, BCKB, DOUT1, DOUT2 — ADC_I2S.SCKA, ADC_I2S.LRCKA, ADC_I2S.BCKA, ADC_I2S.SCKBB, ADC_I2S.LRCKB, ADC_I2S.BCKB, ADC_I2S.DOUT1, ADC_I2S.DOUT2

INT_SIG — INTA, INTB, INTC, INTA#, INTB#, INTC# — INTA, INTB, INTC, INTA#, INTB#, INTC#

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Date: 2/7/2023	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 11

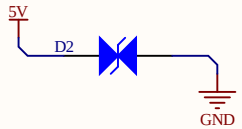
PROGRAM INPUTS



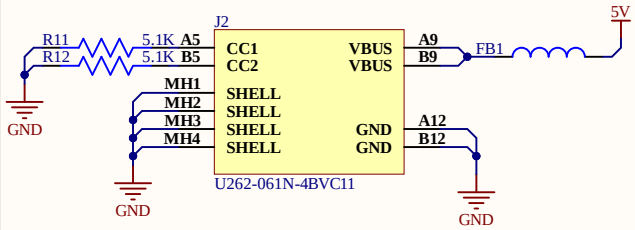
DEBUGGING SIGNALS



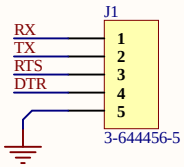
ESD PROTECTION



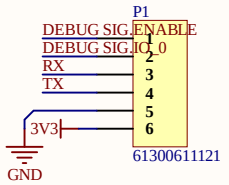
5V INPUT



PROGRAMMING HEADER



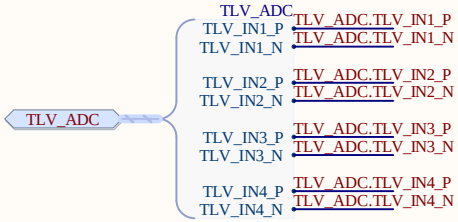
DEBUG CONNECTOR

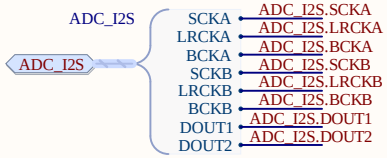


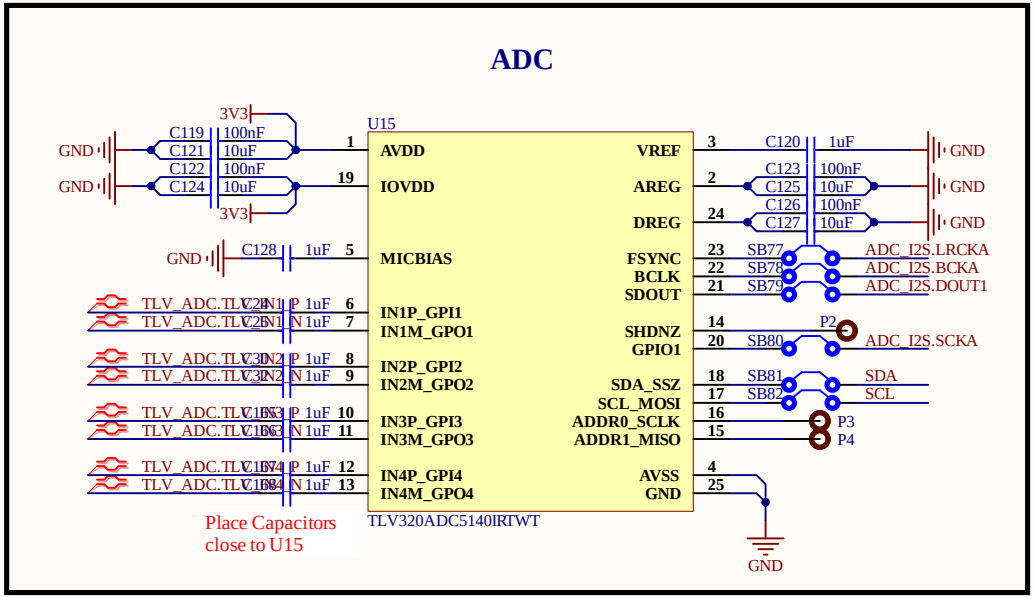


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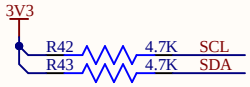
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Date: 2/7/2023	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 11







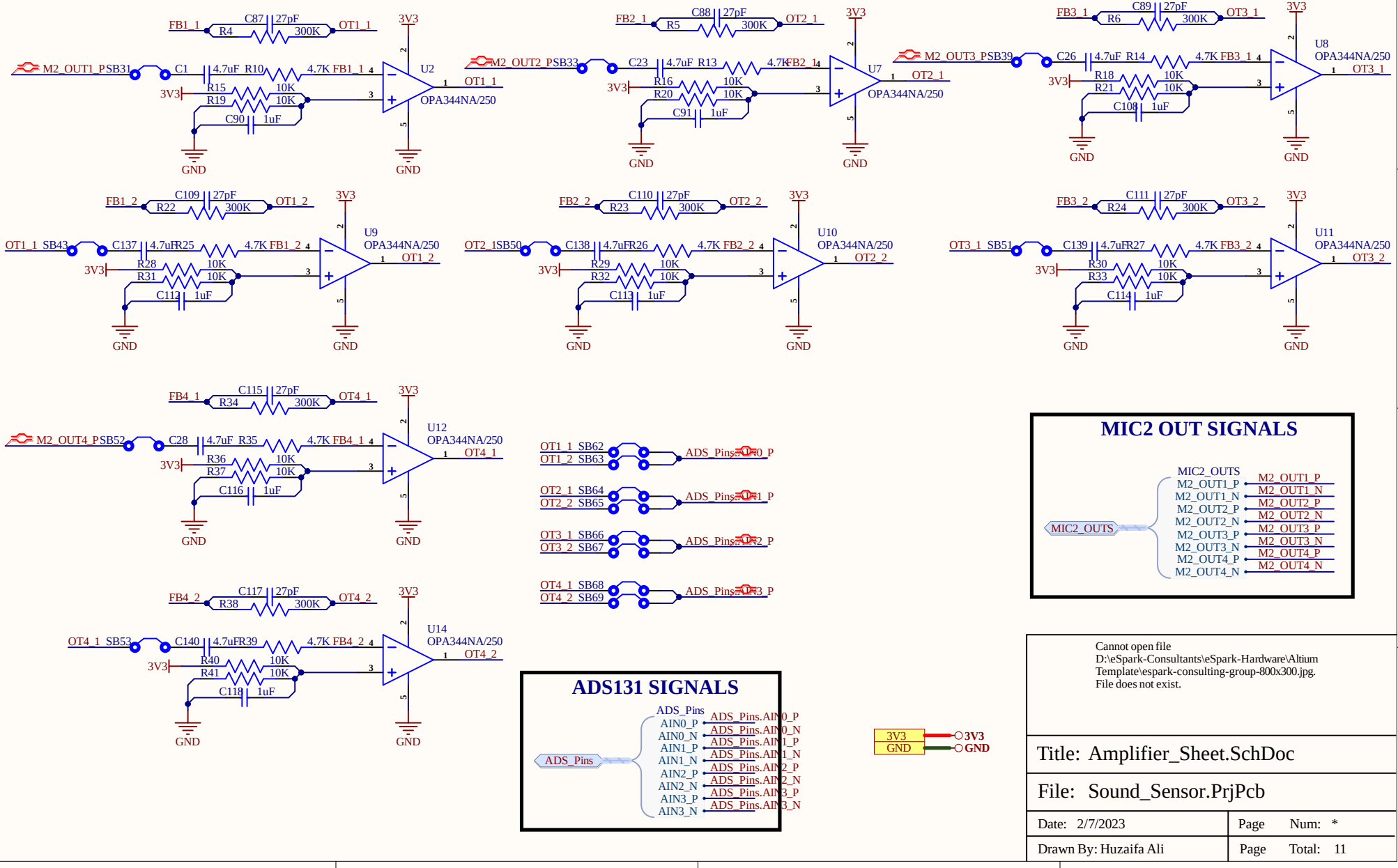




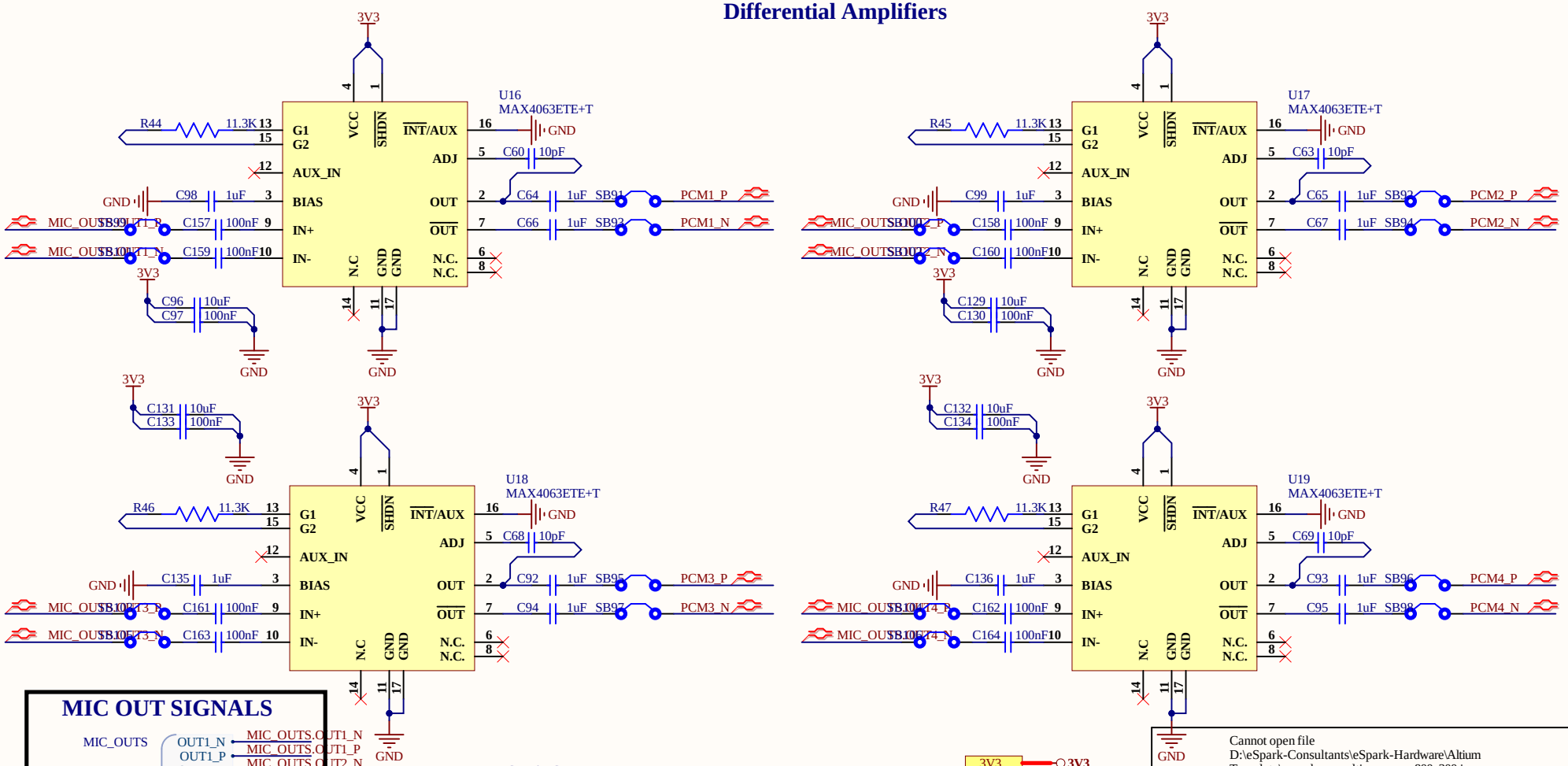


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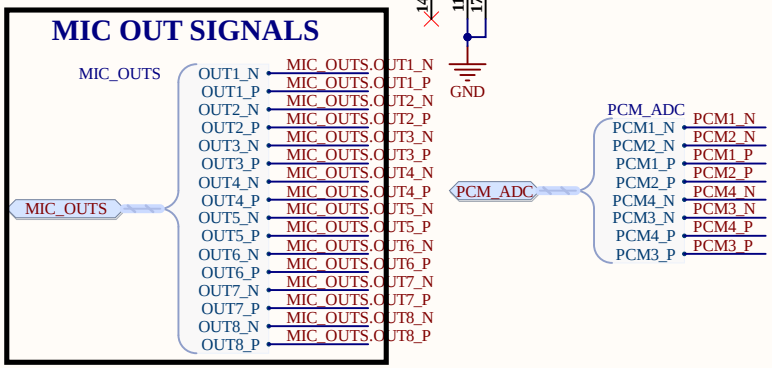
Title: ADC_3_Section.SchDoc	
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Date: 2/7/2023	Page Num: *
Drawn By: Huzaifa Ali	Page Total: 11


FB1_1
C87 27pF
R4 300K
OT1_1
3V3
M2_OUT1_PSB31
C1 4.7uF
R10 4.7K
FB1_1
U2
R15 10K
R19 10K
C90 1uF
OPA344NA/250
GND
FB2_1
C88 27pF
R5 300K
OT2_1
M2_OUT2_PSB33
C23 4.7uF
R13 4.7K
FB2_1
U7
R16 10K
R20 10K
C91 1uF
OPA344NA/250
FB3_1
C89 27pF
R6 300K
OT3_1
M2_OUT3_PSB39
C26 4.7uF
R14 4.7K
FB3_1
U8
R18 10K
R21 10K
C108 1uF
OPA344NA/250
FB1_2
C109 27pF
R22 300K
OT1_2
OT1_1_SB43
C137 4.7uF
R25 4.7K
FB1_2
U9
R28 10K
R31 10K
C112 1uF
OPA344NA/250
FB2_2
C110 27pF
R23 300K
OT2_2
OT2_1SB50
C138 4.7uF
R26 4.7K
FB2_2
U10
R29 10K
R32 10K
C113 1uF
OPA344NA/250
FB3_2
C111 27pF
R24 300K
OT3_2
OT3_1_SB51
C139 4.7uF
R27 4.7K
FB3_2
U11
R30 10K
R33 10K
C114 1uF
OPA344NA/250
FB4_1
C115 27pF
R34 300K
OT4_1
M2_OUT4_PSB52
C28 4.7uF
R35 4.7K
FB4_1
U12
R36 10K
R37 10K
C116 1uF
OPA344NA/250
OT1_1_SB62
OT1_2_SB63
ADS_Pins.AIN0_P
OT2_1_SB64
OT2_2_SB65
ADS_Pins.AIN1_P
OT3_1_SB66
OT3_2_SB67
ADS_Pins.AIN2_P
OT4_1_SB68
OT4_2_SB69
ADS_Pins.AIN3_P
FB4_2
C117 27pF
R38 300K
OT4_2
OT4_1_SB53
C140 4.7uF
R39 4.7K
FB4_2
U14
R40 10K
R41 10K
C118 1uF
OPA344NA/250
MIC2 OUT SIGNALS
MIC2_OUTS
M2_OUT1_P
M2_OUT1_N
M2_OUT2_P
M2_OUT2_N
M2_OUT3_P
M2_OUT3_N
M2_OUT4_P
M2_OUT4_N
ADS131 SIGNALS
ADS_Pins
AIN0_P
AIN0_N
AIN1_P
AIN1_N
AIN2_P
AIN2_N
AIN3_P
AIN3_N
3V3
GND
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Title: Amplifier_Sheet.SchDoc
File: Sound_Sensor.PrjPcb
Date: 2/7/2023
Page Num: *
Drawn By: Huzaifa Ali
Page Total: 11

Differential Amplifiers



MIC OUT SIGNALS

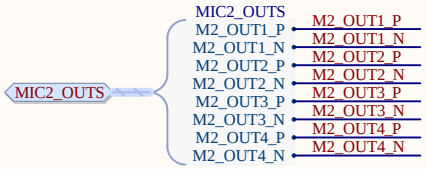




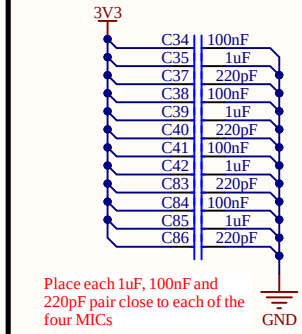
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Title: Diff_Amplifier_Sheet.SchDoc	
File: Sound_Sensor.PrjPcb	
Date: 2/7/2023	Page Num: *
Drawn By: Huzaifa Ali	Page Total: 11

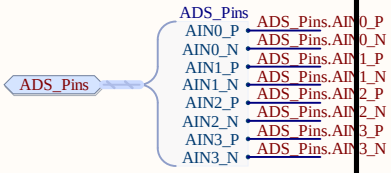
MIC2 OUT SIGNALS



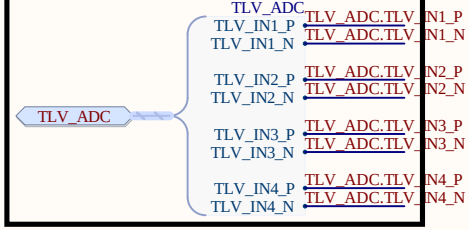
DECOUPLING CAPS



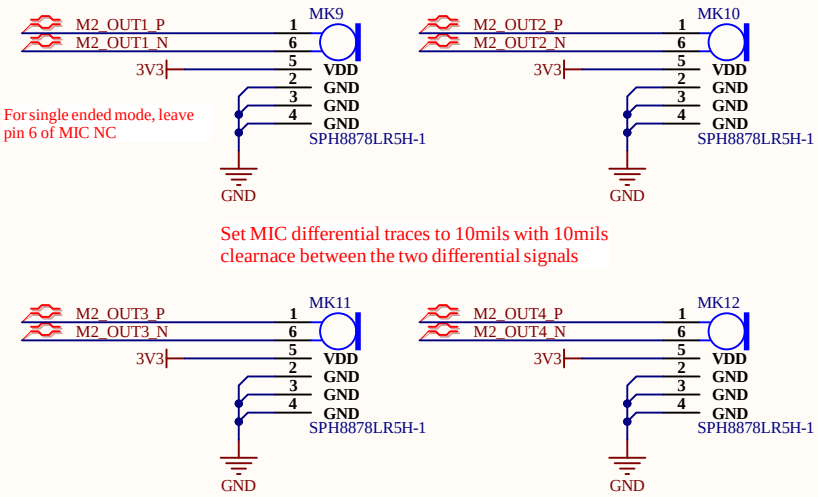
ADS131 SIGNALS



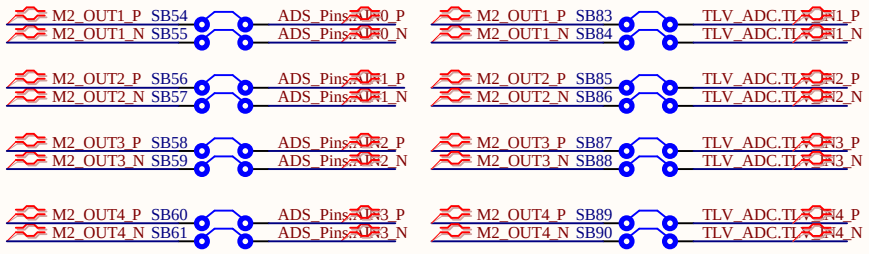
TLV320ADC5140



ANALOG MIC2 INPUTS



Selection





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Title: Sensor_Sheet_2.SchDoc	
File: Sound_Sensor.PrjPcb	
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Drawn By: Huzaifa Ali	Page Total: 11